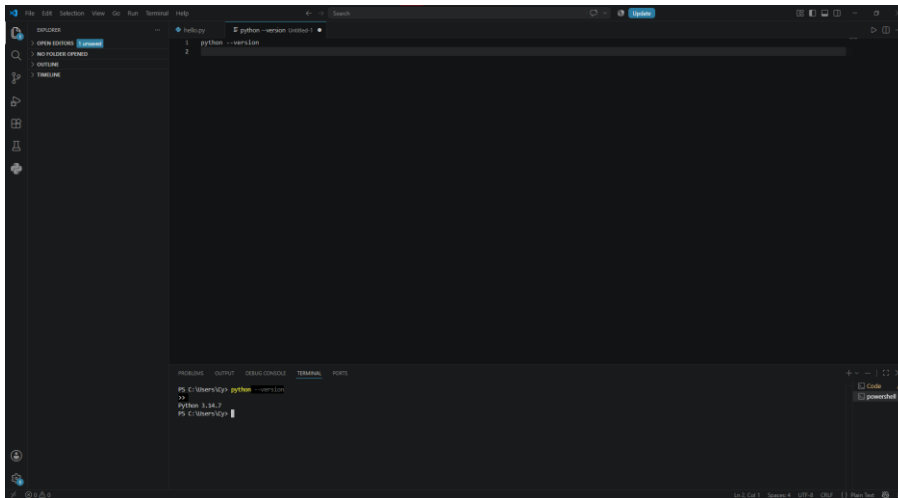


1. The terminal showing **python --version** and its answer.



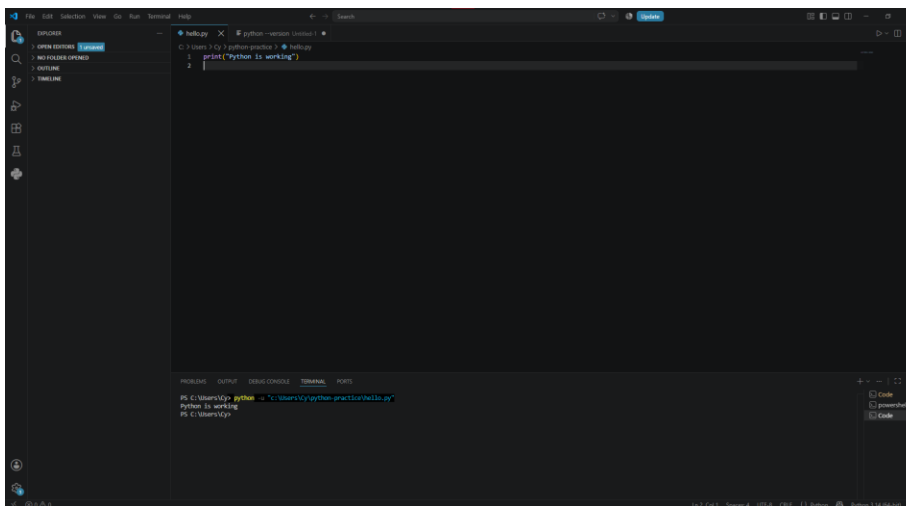
```
PS C:\Users\Cy> python --version
```

```
>>
```

```
Python 3.14.7
```

```
PS C:\Users\Cy>
```

2. VS Code with **hello.py** open and its output visible in the terminal.

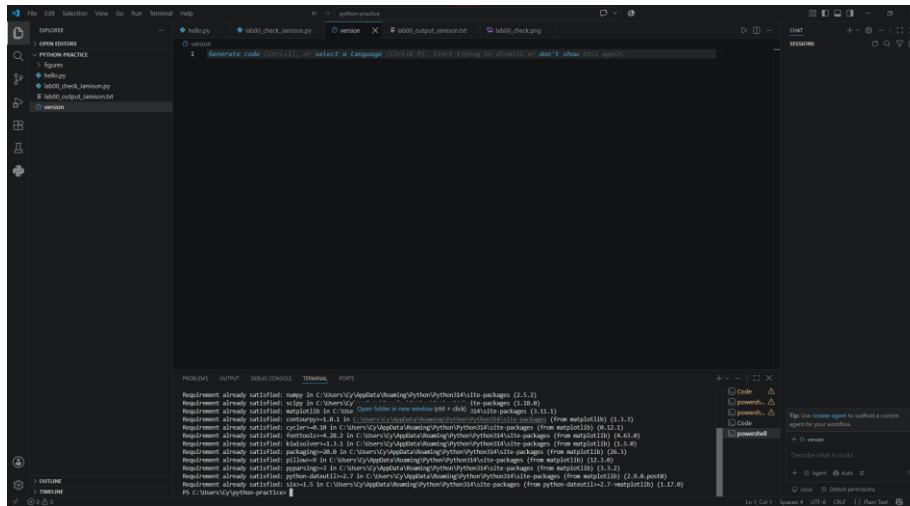


```
PS C:\Users\Cy\python-practice> python -u "c:\Users\Cy\python-practice\hello.py"
```

```
Python is working
```

```
PS C:\Users\Cy\python-practice>
```

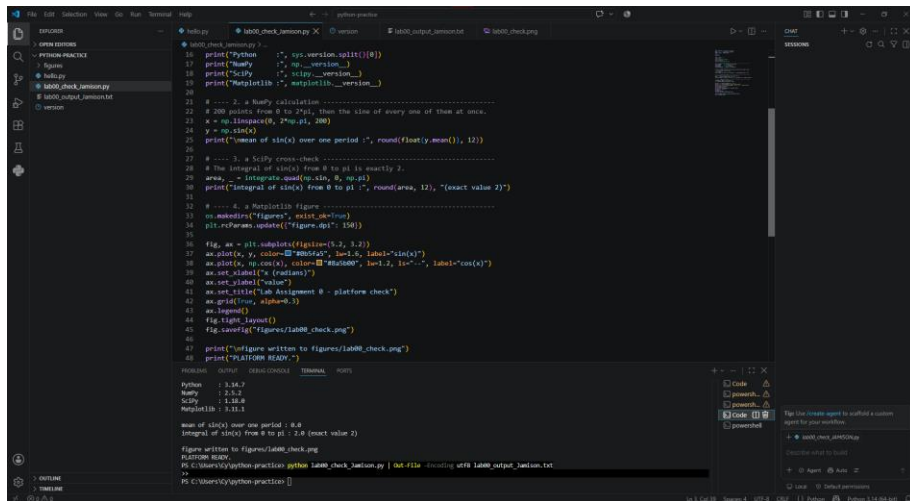
3. The terminal showing the successful installation of NumPy, SciPy and Matplotlib.



```
1 Generate code (ctrl+alt+L) or select a language (ctrl+alt+J) to start typing or don't show this message.

Requirements already satisfied: numpy in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (1.24.2)
Requirements already satisfied: scipy in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (1.10.4)
Requirements already satisfied: matplotlib in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (3.8.0)
Requirements already satisfied: contourpy in 4.0.1 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (3.3.3)
Requirements already satisfied: cycler in 0.12.1 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (0.12.1)
Requirements already satisfied: fonttools in 4.22.1 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (4.22.1)
Requirements already satisfied: kiwisolver in 3.0.13 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (3.0.13)
Requirements already satisfied: packaging in 24.0 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (24.0)
Requirements already satisfied: pillow in 10.4.0 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (10.4.0)
Requirements already satisfied: pyparsing in 3.1.2 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (3.1.2)
Requirements already satisfied: python-dateutil in 2.9.0 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (2.9.0)
Requirements already satisfied: six in 1.16.0 in C:\Users\j\AppData\Local\Programs\Python\Python310\Scripts (from matplotlib) (1.16.0)
PS C:\Users\j\python-practice
```

4. The full console output of your verification program, ending in PLATFORM READY.



```
16 print("Python", sys.version.split()[0])
17 print("NumPy", np.__version__)
18 print("SciPy", scipy.__version__)
19 print("Matplotlib", matplotlib.__version__)
20
21 # --- 2. a NumPy calculation ---
22 # 200 points from 0 to 2*pi, then the size of every one of them at once.
23 x = np.linspace(0, 2*np.pi, 200)
24 y = np.sin(x)
25 print("Mean of sin(x) over one period:", round(float(y.mean()), 12))
26
27 # --- 3. a SciPy cross-check ---
28 # The integral of sin(x) from 0 to pi is exactly 2.
29 area, _ = integrate.quad(np.sin, 0, np.pi)
30 print("Integral of sin(x) from 0 to pi:", round(area, 12), "(exact value 2)")
31
32 # --- 4. a Matplotlib figure ---
33 plt.figure(figsize=(5, 3))
34 plt.clf()
35 plt.plot(x, y, color="blue", label="sin(x)")
36 plt.plot(x, np.cos(x), color="red", label="cos(x)")
37 plt.xlabel("x (radians)")
38 plt.ylabel("y (value)")
39 plt.title("Lab assignment 0 - platform check")
40 plt.grid(True, alpha=0.3)
41 plt.legend()
42 fig.savefig("figures/lab0-check.png")
43
44 print("Figure written to figures/lab0-check.png")
45 print("PLATFORM READY.")
```

```
Python: 3.10.2
NumPy: 1.24.2
SciPy: 1.10.4
Matplotlib: 3.8.0
Mean of sin(x) over one period: 0.0
Integral of sin(x) from 0 to pi: 2.0 (exact value 2)
Figure written to figures/lab0-check.png
PLATFORM READY
PS C:\Users\j\python-practice python lab0_check.py -Ouf-file -creating ufb lab0_output_lab0m.txt
```

5. User GitHub repository page, showing the repository name and the uploaded file.

